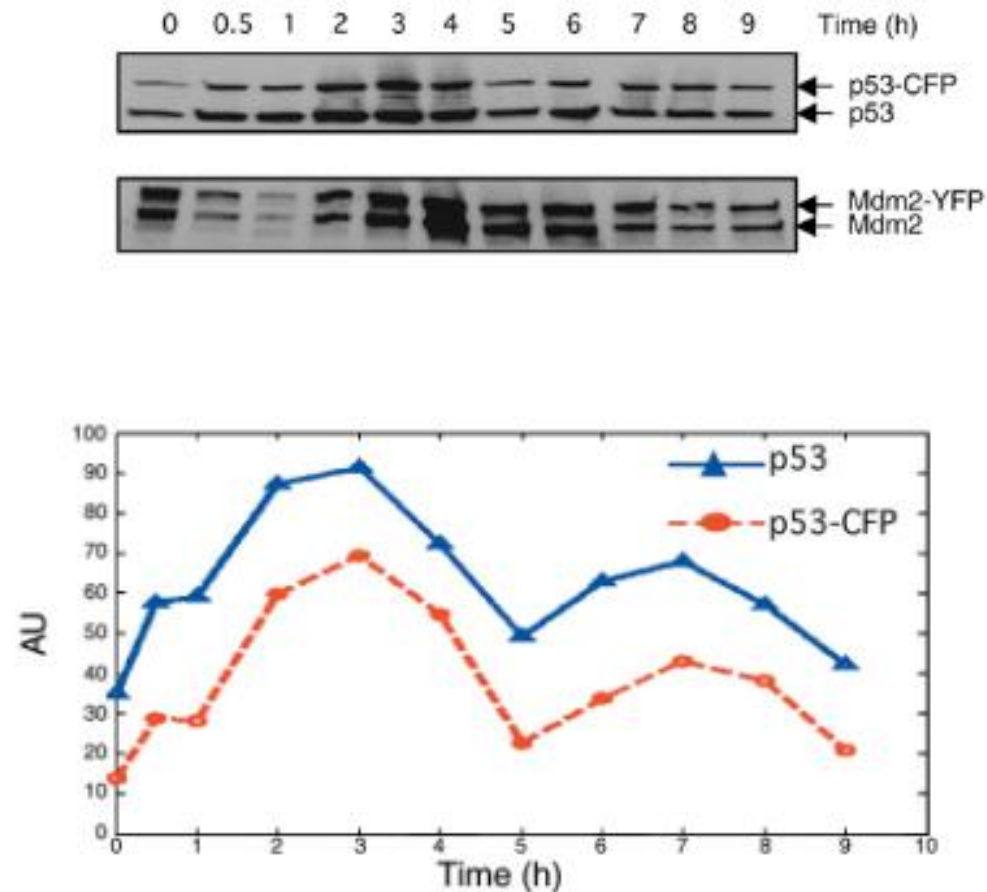


How to build a biological oscillator?

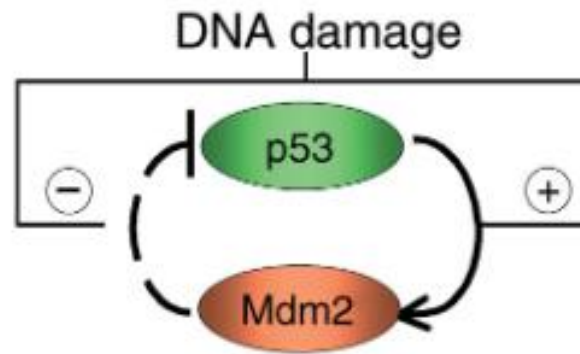
p53-Mdm2 oscillations and Galit Lahav

Population measurements suggested that p53 and Mdm2 undergo damped oscillations after DNA damage

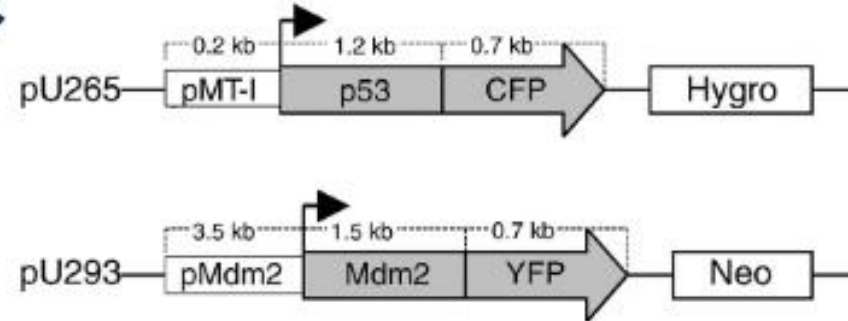


The p53–Mdm2 loop can be followed dynamically in individual living cells

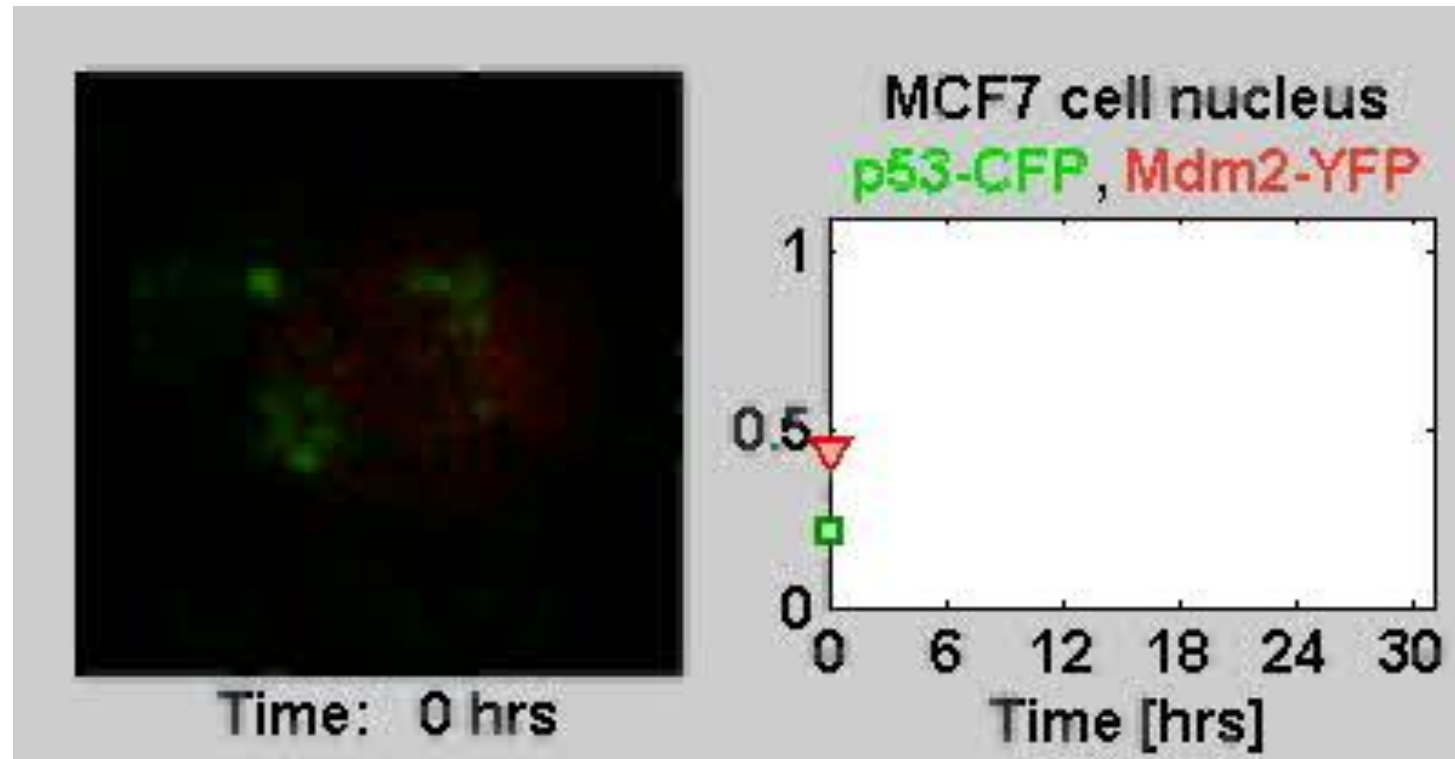
b



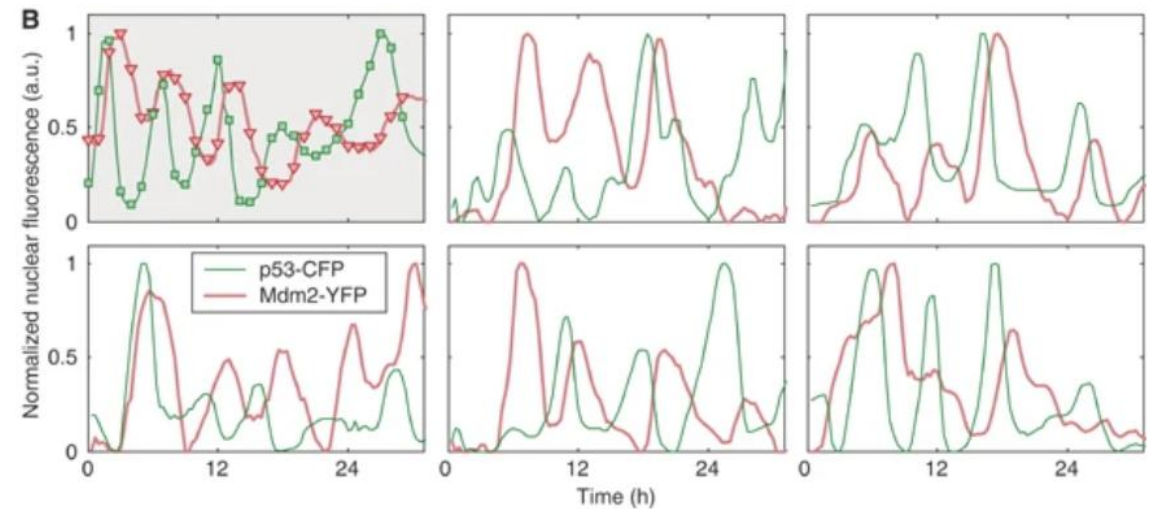
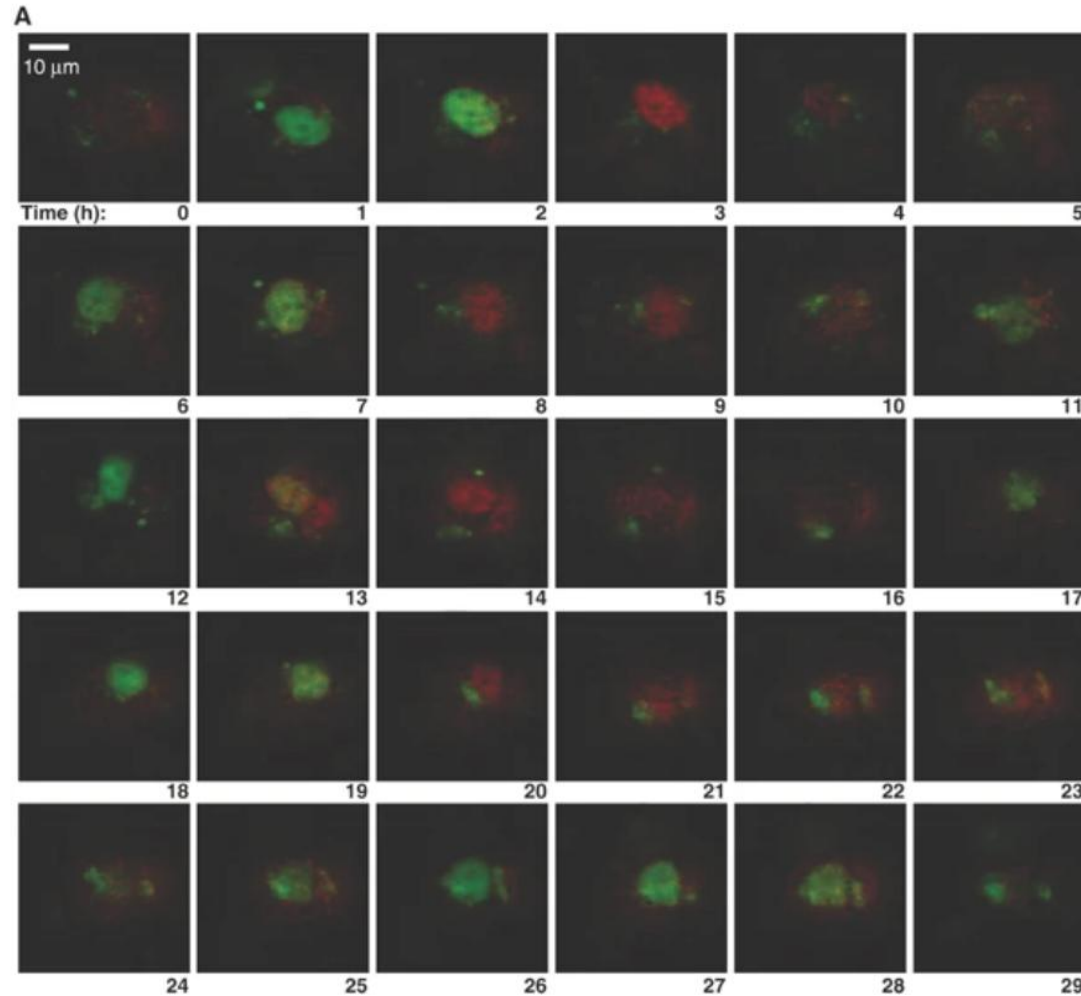
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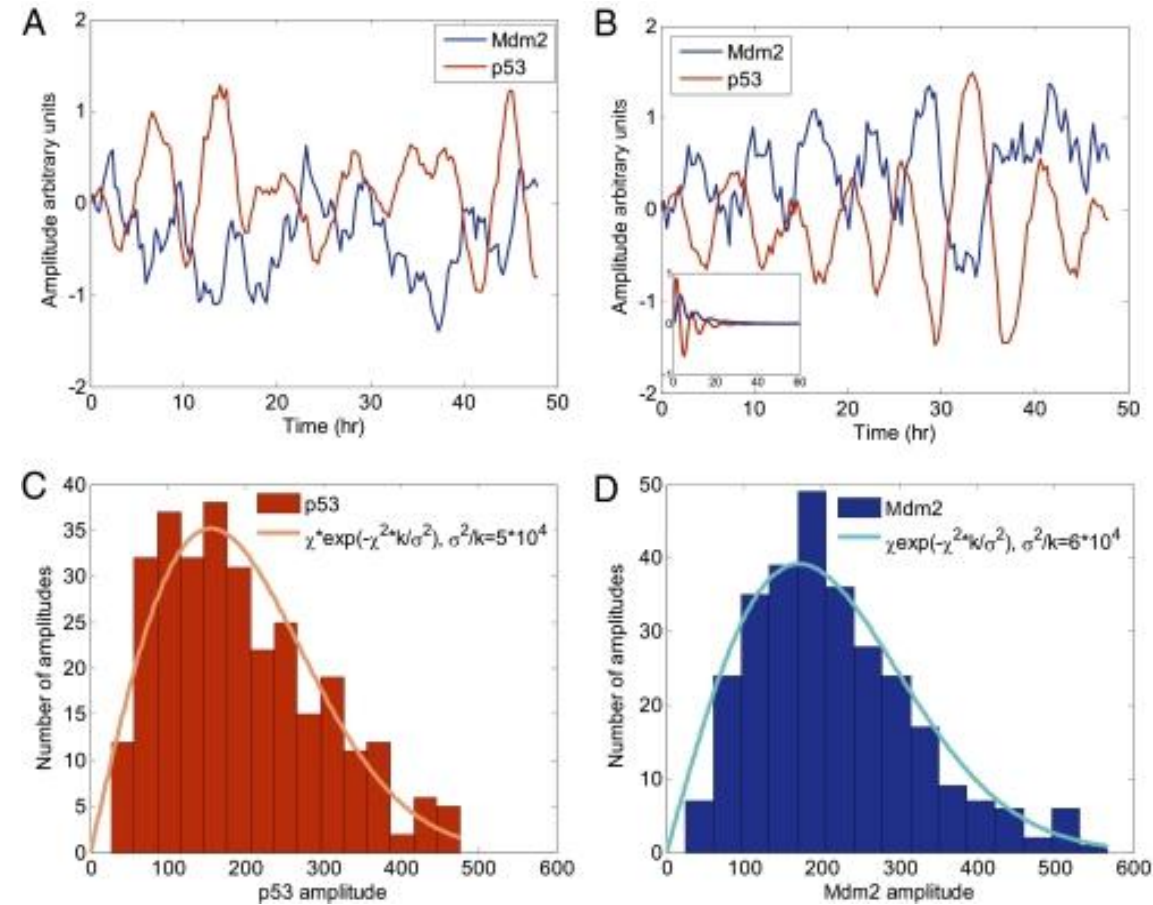
In single cells, the p53–Mdm2 circuit generates sustained oscillations



p53–Mdm2 oscillations are more precise in period than in amplitude



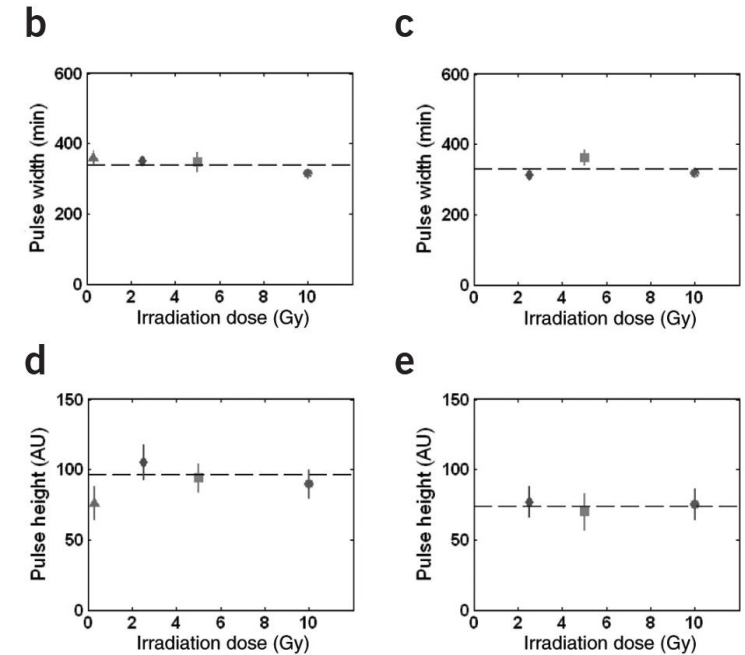
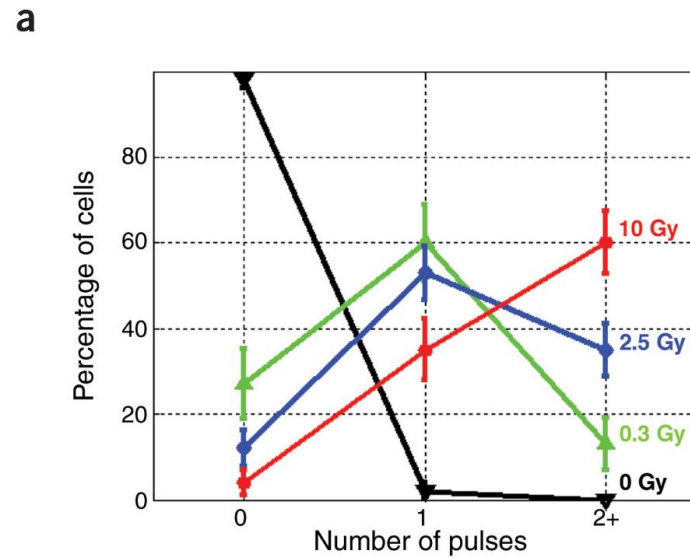
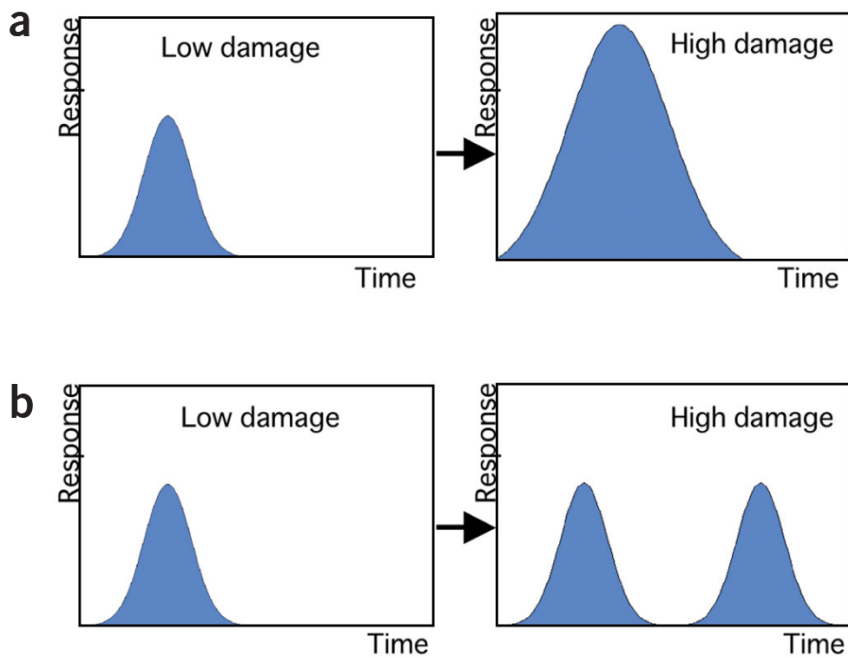
Stochastic noise can sustain oscillations around a stable spiral



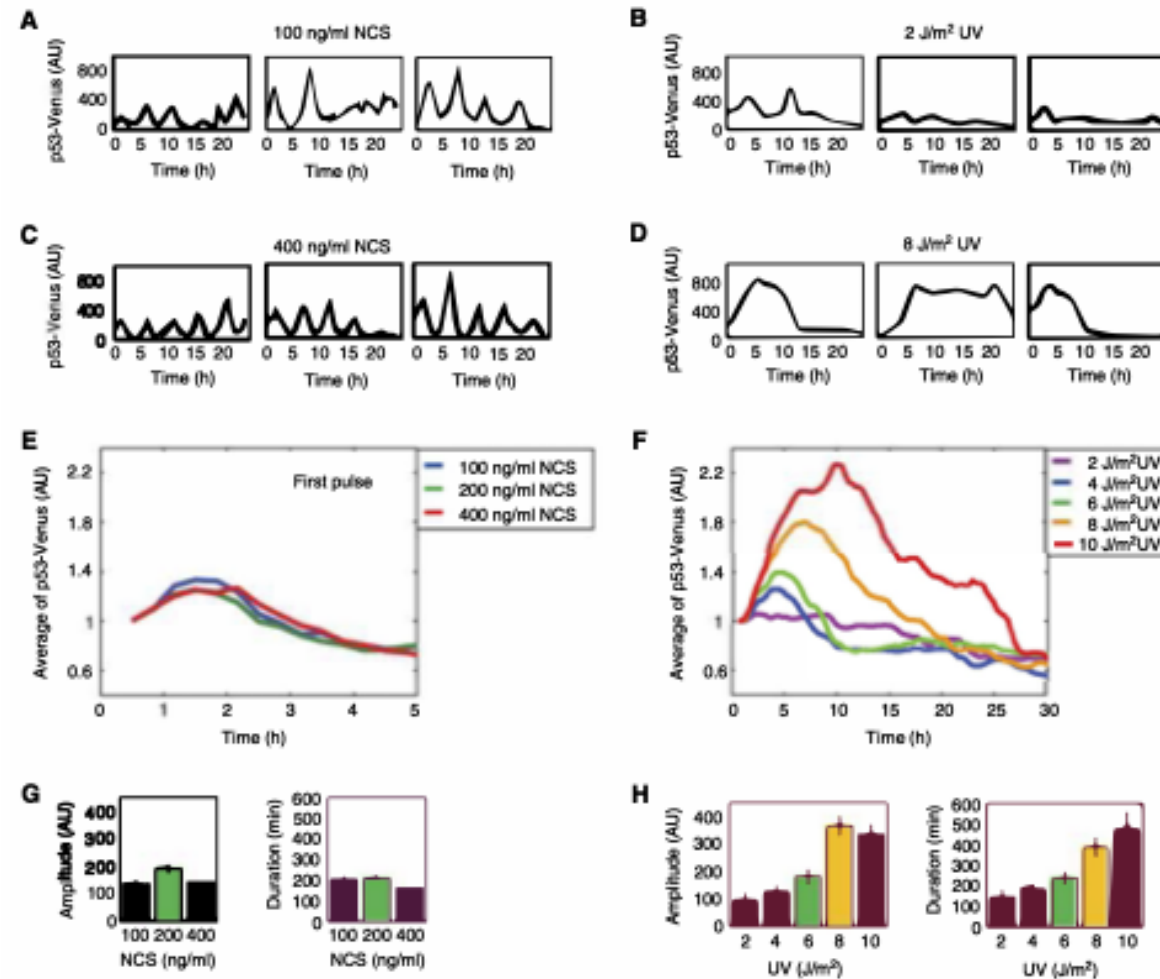
Negative-feedback loops recur across many biological regulatory systems



DNA damage is encoded digitally in the number of p53 pulses, rather than continuously in pulse amplitude

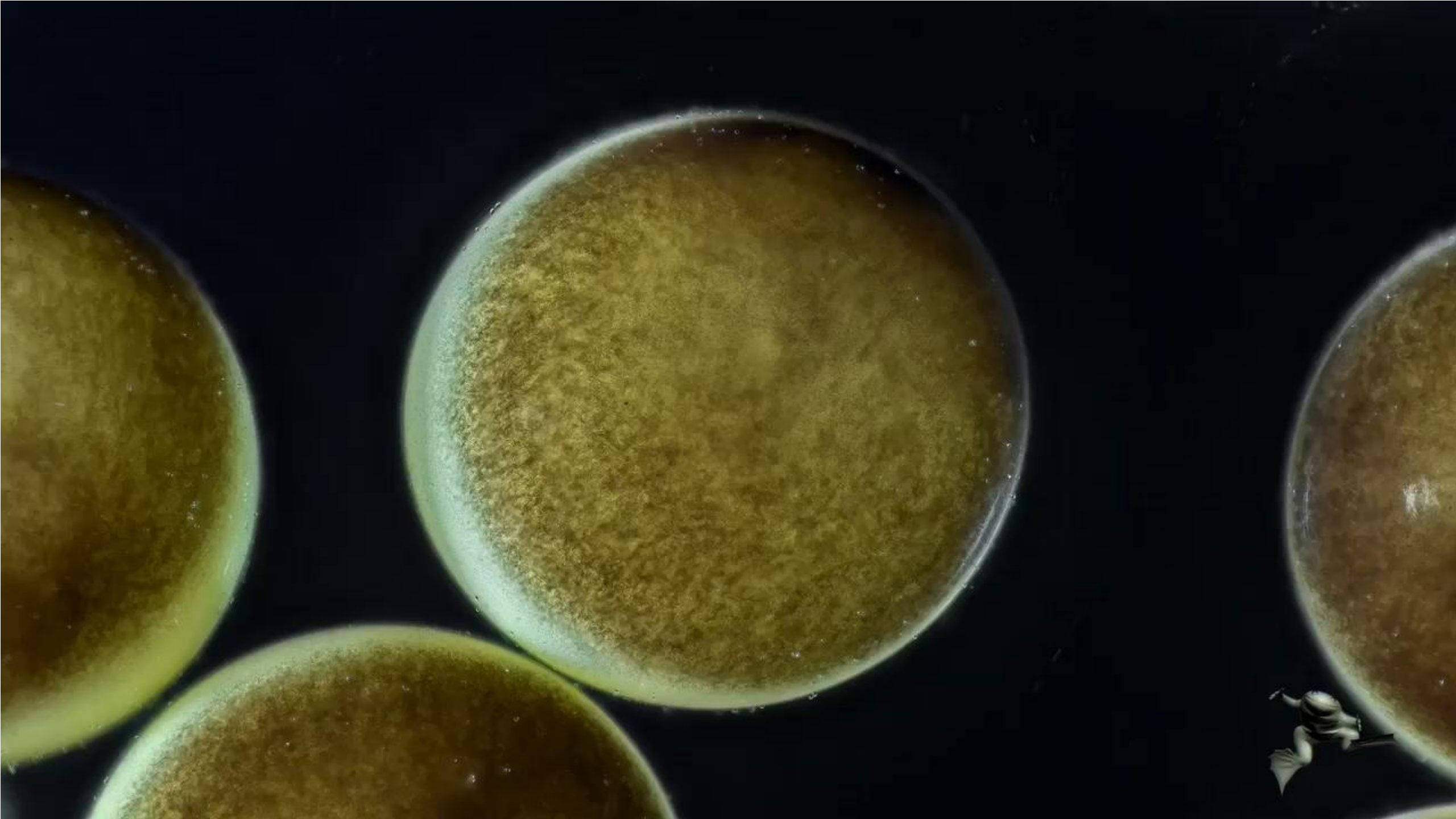


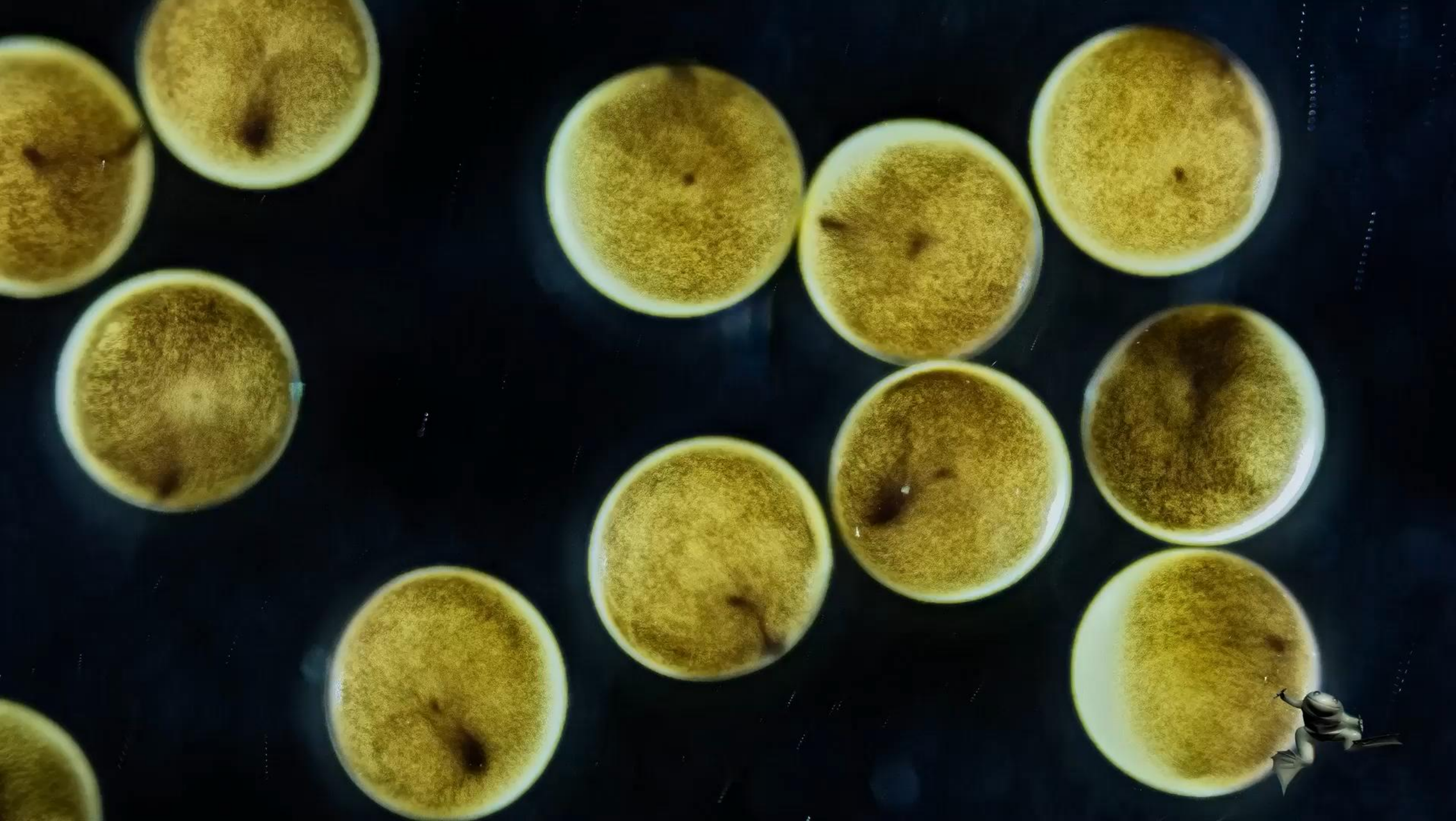
Different forms of DNA damage generate distinct p53 dynamics



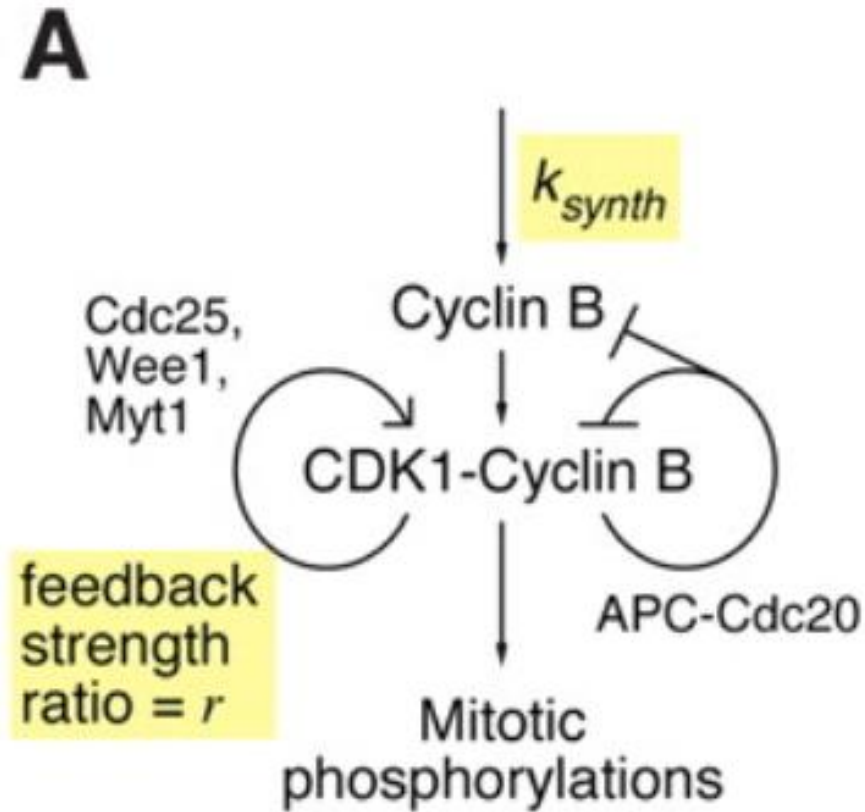
Positive feedback plus negative feedback motif and cell-cycle oscillations

Early *Xenopus* embryo divide periodically in a clock-like fashion

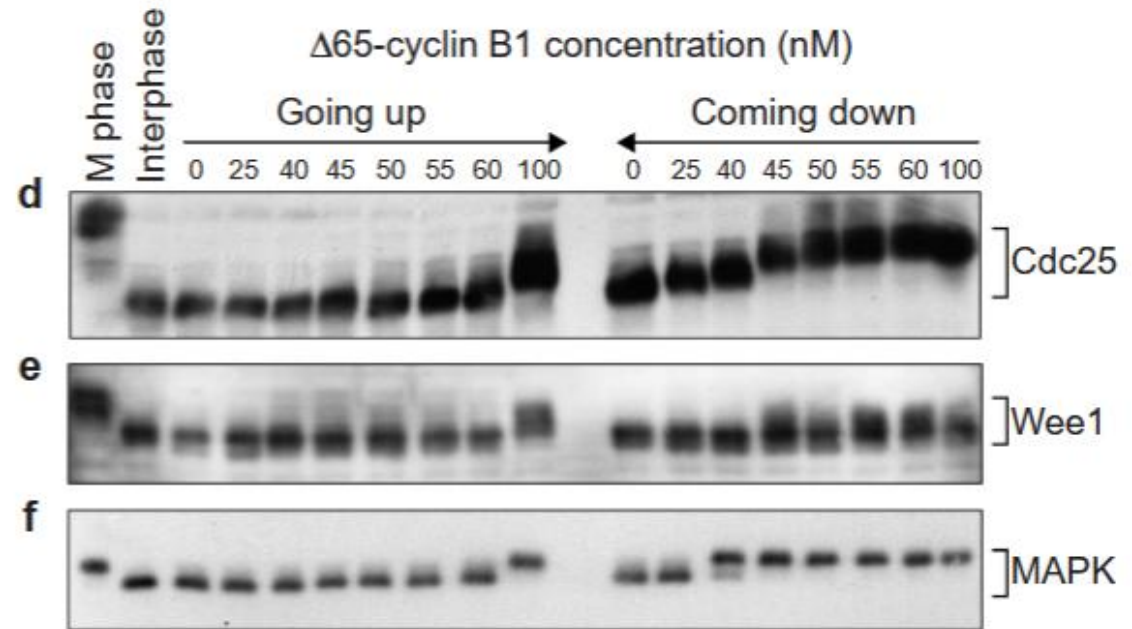
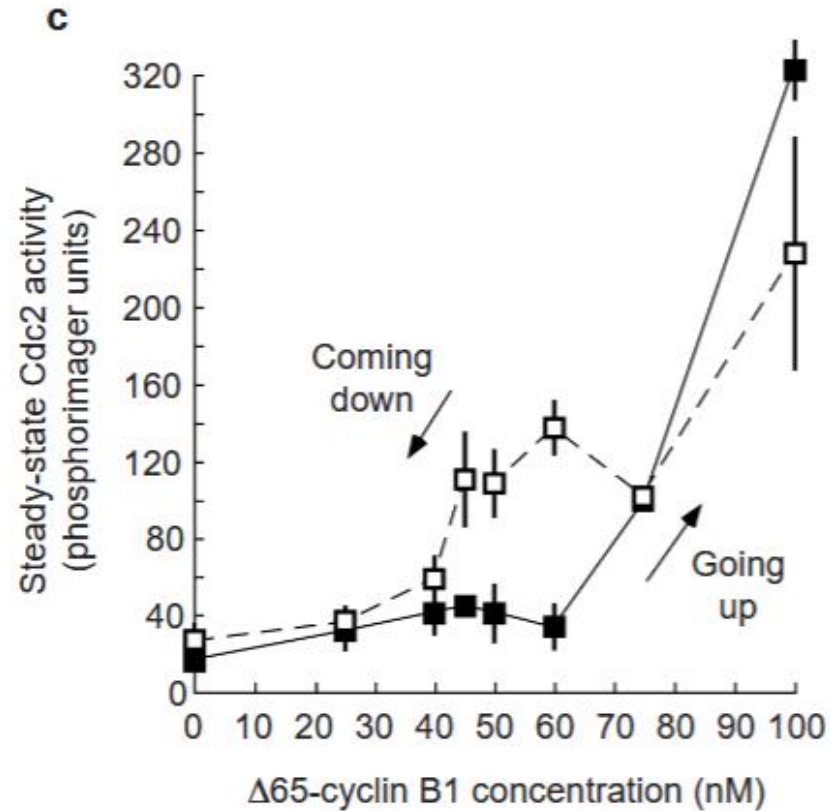




Early embryonic cell cycle is driven by a positive plus negative feedback motif between CDK1-CyclinB and APC-Cdc20



Hysteresis makes oscillation even more possible



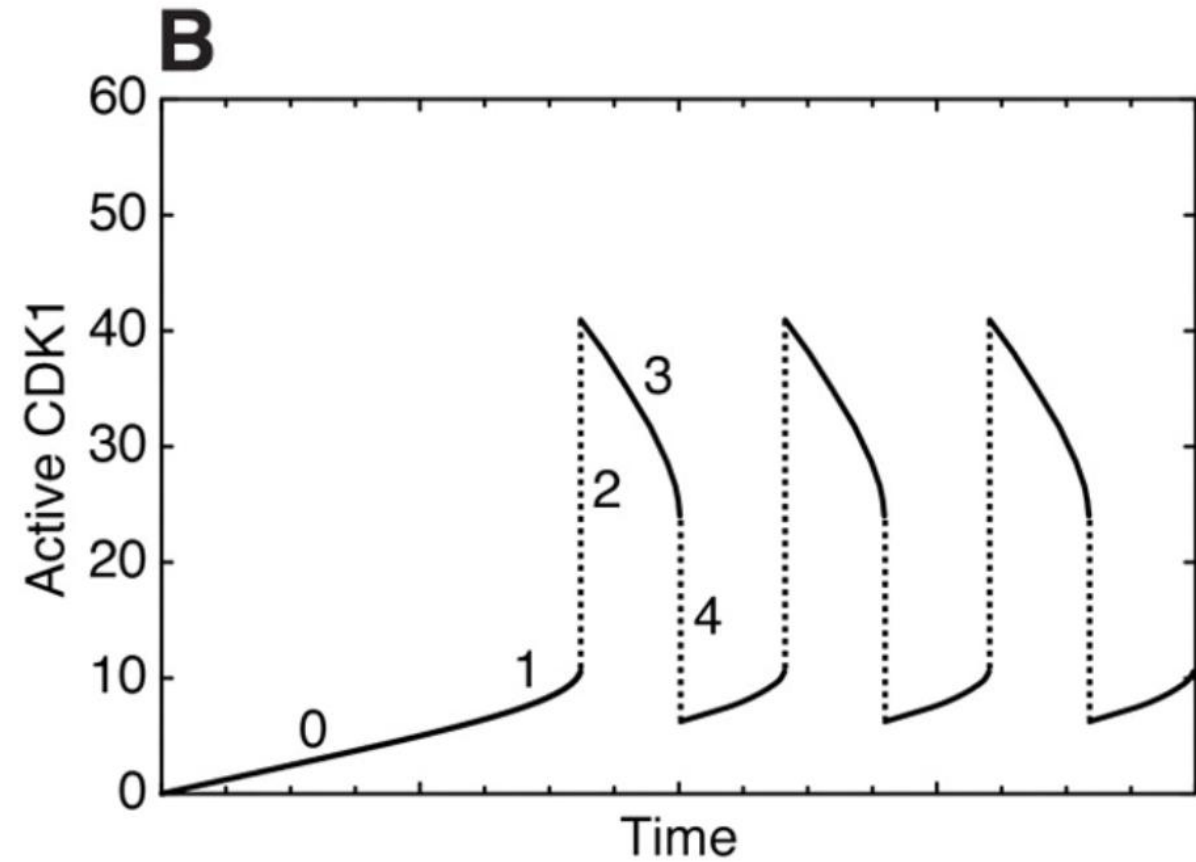
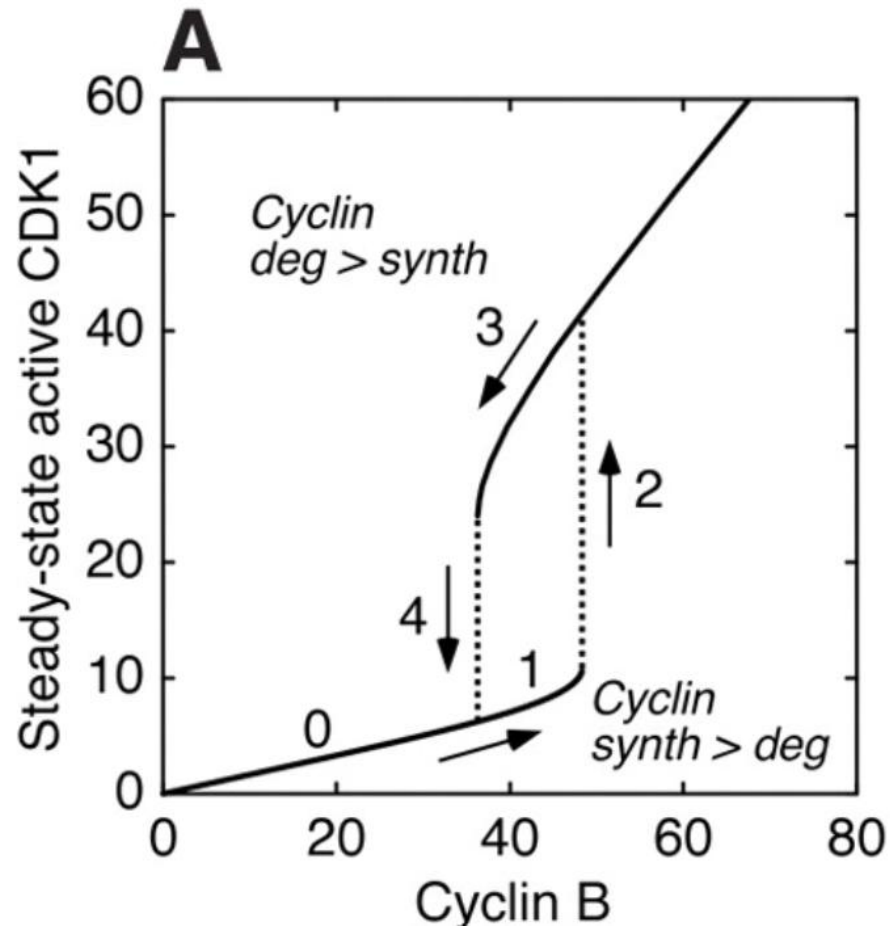
This model of biological oscillations was developed by an interdisciplinary team including Chao Tang & James Ferrell

Robust, Tunable Biological Oscillations from Interlinked Positive and Negative Feedback Loops

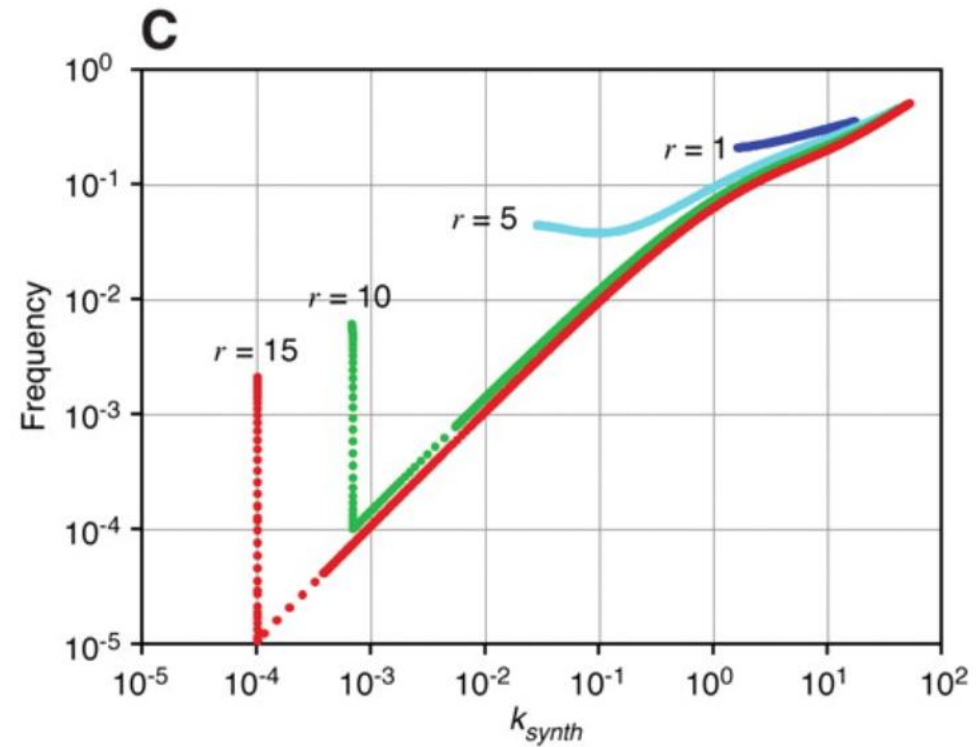
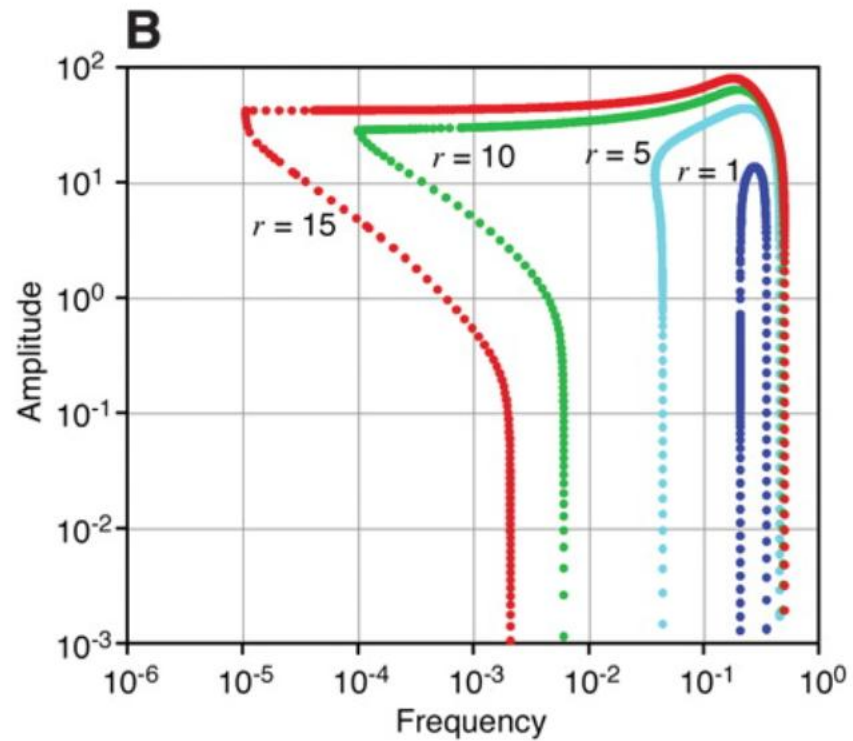
Tony Yu-Chen Tsai,^{1*} Yoon Sup Choi,^{1,2*} Wenzhe Ma,^{3,4} Joseph R. Pomeroy,⁵ Chao Tang,^{3,4} James E. Ferrell Jr.^{1†}

A simple negative feedback loop of interacting genes or proteins has the potential to generate sustained oscillations. However, many biological oscillators also have a positive feedback loop, raising the question of what advantages the extra loop imparts. Through computational studies, we show that it is generally difficult to adjust a negative feedback oscillator's frequency without compromising its amplitude, whereas with positive-plus-negative feedback, one can achieve a widely tunable frequency and near-constant amplitude. This tunability makes the latter design suitable for biological rhythms like heartbeats and cell cycles that need to provide a constant output over a range of frequencies. Positive-plus-negative oscillators also appear to be more robust and easier to evolve, rationalizing why they are found in contexts where an adjustable frequency is unimportant.

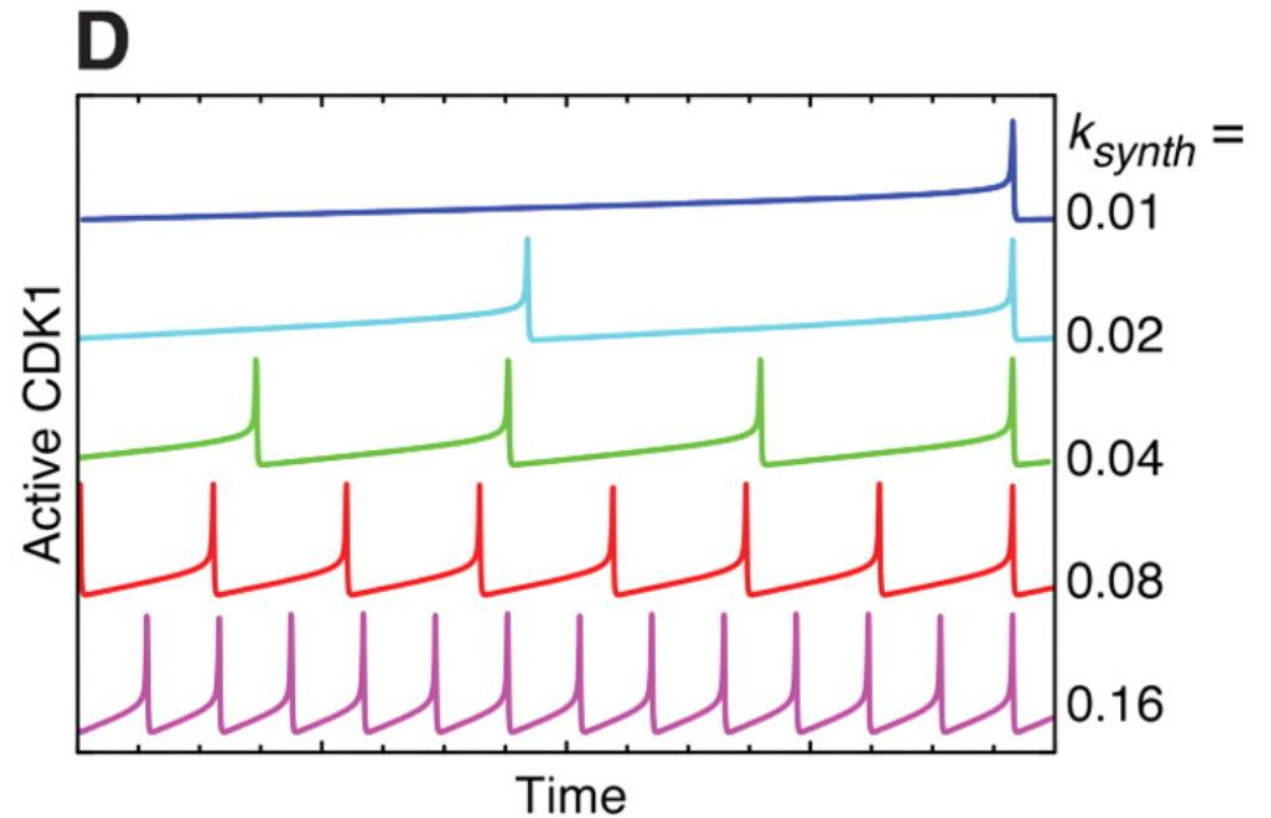
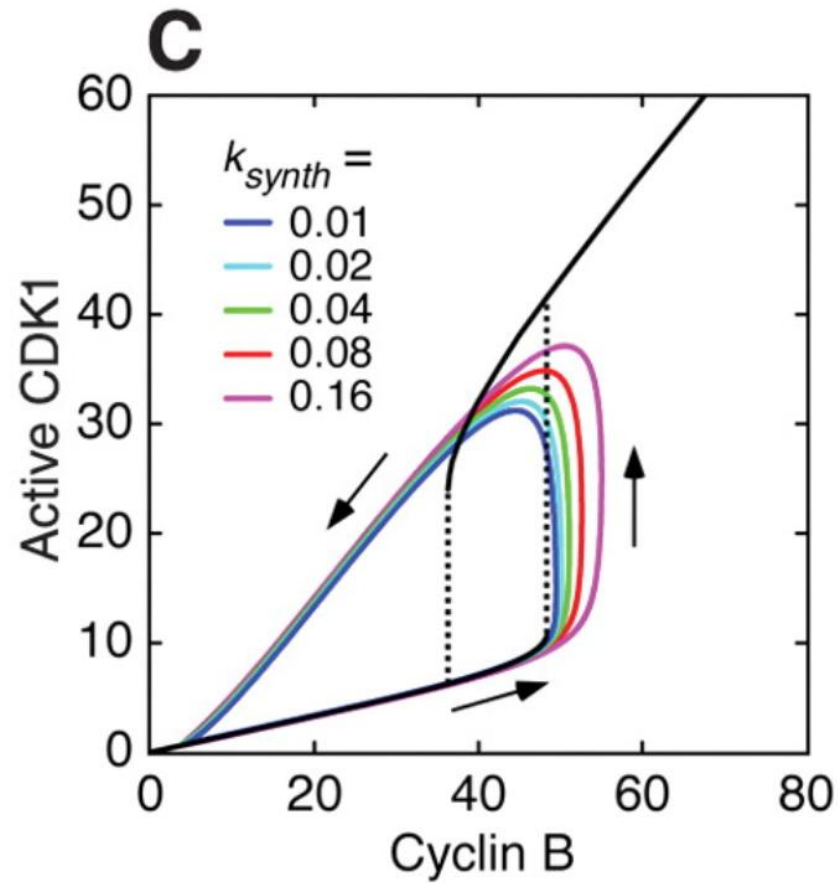
Hysteresis and bistability of CDK1 lead to a relaxation oscillator



Oscillation is made possible by the steepness of positive feedback



Relaxation oscillator's frequency can be tuned by one parameter



Reciprocal feedback on opposing enzymes makes bistability robust

